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# Rules of the Matrix

## Safety & Guardrails

every agent runs inside something — that something is the harness

# The Harness

An agent is a model in a loop. Left alone, it can touch your **files**, your **network**, your **shell**.

A **harness** is the runtime container that enforces *what it can and cannot do*.

You already built one today. In L4, the `ValidateSet` allow-list refused `lab-dc01` — the agent asked, the schema said no.

**"The LLM decides WHEN to call. You decide WHAT is callable."**

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# One Idea, Three Scales

LAYER 1
### Local sandbox
On your machine. Restricts filesystem, network, shell. In your seat.

LAYER 2
### Cloud sandbox
Ephemeral, fully isolated Linux. Off your machine entirely.

LAYER 3
### Enterprise harness
Platform-managed identity & governance. Where this scales.

Same harness idea — increasing control over the blast radius.

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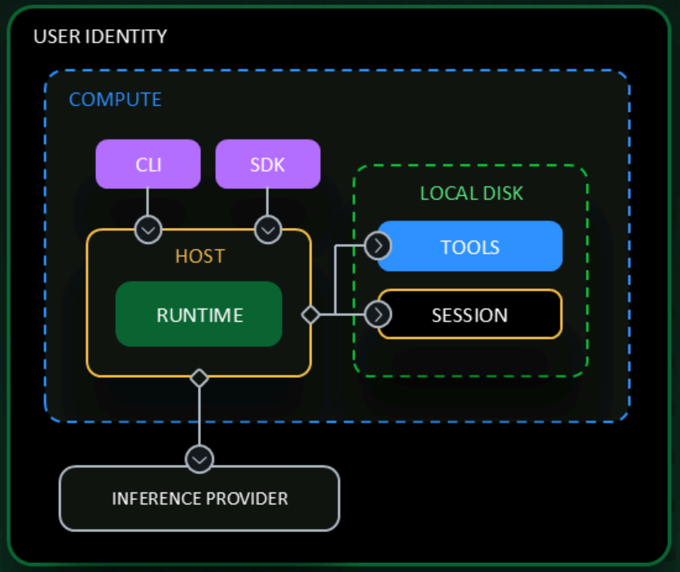
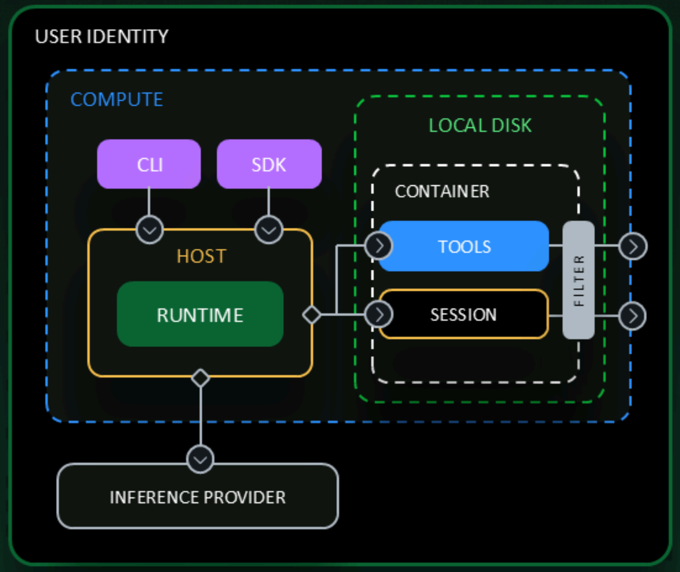
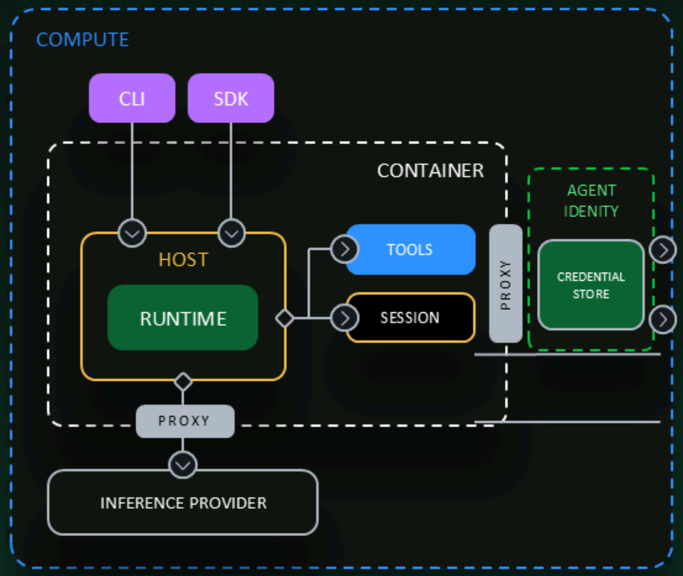
# Three-Tier Sandbox Model

PUBLIC PREVIEW

## Copilot Sandboxes




### No sandbox

POWER USER

The agent runs directly on the user's machine with full access to the file system, network, and installed tools.

### Sandboxed

SAFE TOOL EXECUTION

The main agent loop runs on the host with low permissions, but delegates tool execution to isolated containers.

### Fully isolated

PLATFORM SERVICE

Agents executing within the sandbox have minimal to no knowledge of the resources and identities being used.

No sandbox → Sandboxed → Fully isolated · Source: Microsoft Build FY26

More isolation → less the agent can see. Choose the tier per task.

# Local Sandbox

Microsoft MXC restricts the three things that matter: filesystem, network, shell.

```
> /sandbox enable
# agent loop now runs contained — fs / network / shell limited
> /sandbox disable
# back to full machine access when you need it
```

Included in your seat — no separate meter. For agent work on a laptop, this is the default you want on.

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# Cloud Sandbox & Policy

```
> copilot --cloud
# ephemeral, fully-isolated Linux on GitHub
```

- Frees local resources — runs off your machine
- Resume sessions across devices
- Separate usage meter from your seat

### MDM enforcement

Enterprises centrally enforce sandbox policies through MDM platforms.

You don't rely on every engineer remembering to toggle it — IT enforces it.

# Beyond the Sandbox: The Enterprise Harness

### Azure AI Foundry

Build, evaluate, and run agents with governance, observability, and content safety built in — the fully-isolated tier, operationalized.

### Microsoft Agent 365

Agents as first-class, governed identities in your Entra estate: registry, access control, lifecycle.

The harness you hand-built with `ValidateSet` — this is where it becomes a managed platform at org scale. Today we just show the runway.

# Bonus: The Desktop App Is a Harness Too

GitHub Copilot **Desktop App** runs each session in its own isolated **git worktree**.

- Parallel agents can't step on each other's working tree
- Isolation by construction, not by discipline

Same harness instinct, at the workspace level.

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# What DNV Can Do Tomorrow

### 1 · Enable sandboxes by default

Make `/sandbox enable` the team norm. Full machine access becomes the exception you opt into.

### 2 · Ship an MDM policy starter

IT enforces sandbox policy centrally — safety stops depending on per-engineer memory.

### 3 · Reuse the L4 scoped-agent pattern

`ValidateSet` allow-lists, read-only by default. Terraform tools read state & plan; gate `apply` behind a human.

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# Rules of the Matrix

Local sandbox → cloud sandbox → enterprise harness.

**Give the agent power inside a harness — and the harness is yours to design.**

Read more: GitHub Changelog (2026-06-02) · docs.github.com/copilot/concepts/about-cloud-and-local-sandboxes · Build DEM305